

Less Is More: A Stealthy and Efficient Adversarial Attack Method for DRL-Based Autonomous Driving Policies

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Abstract—Existing research has demonstrated that autonomous driving policies based on deep reinforcement learning (DRL) are vulnerable to adversarial attacks, which poses challenges for the practical deployment of these policies. Designing effective adversarial attacks is a crucial prerequisite for building robust driving policies. In view of this, we propose a novel adversarial attack method, which can attack the DRL-based autonomous driving agents in a stealthy and efficient manner. This method models the attack as a mixed-integer optimization problem that aims to maximize the safety violations (e.g., collisions) of the agents while minimizing the number of attack steps. Then, a DRL-based adversary is devised in this method to solve the problem to automatically learn the optimal attack policy without domain knowledge. To further enhance the adversary's learning capability, this method incorporates attack-related information into its observations to provide more decision-making context and employ a trajectory clipping technique to enhance sample quality. Extensive evaluation results reveal that our method achieves a remarkable 105% enhancement in attack efficiency compared to existing methods.

Index Terms—Adversarial attacks, autonomous vehicle, deep reinforcement learning (DRL), robust decision making.

I. INTRODUCTION

IN RECENT years, autonomous driving has achieved remarkable progress, propelled by rapid advancements in sensors and artificial intelligence (AI) [1], [2]. Among the various AI paradigms, deep reinforcement learning (DRL) stands out for its exceptional ability to tackle complex decision-making challenges, particularly in dynamic and uncertain environments. This distinctive capability makes DRL a highly promising approach for advancing the development of more intelligent autonomous driving policies [3]. Given this potential, global technology companies, including Waymo, Tesla, and Baidu Apollo have been adopting DRL in their development pipelines, striving to pioneer next-generation automotive

solutions [4]. Research efforts have further validated DRL-based policies in diverse autonomous driving tasks, including highway driving [5], [6], [7], on-ramp merging [8], [9], [10], and intersection navigation [3], [11], [12].

Despite significant advancements, the vulnerability of DRL to adversarial perturbations remains a critical challenge, hindering its reliable deployment in safety-critical tasks, such as autonomous driving [13], [14]. In real-world driving scenarios, adversarial perturbations can arise from various sources, including sensor noise, environmental interference, and even maliciously crafted inputs designed to deceive the system. These perturbations can significantly degrade the performance of DRL-based policies, leading to erroneous decisions, such as incorrect lane changes.

Given the high-stakes nature of autonomous driving, studying adversarial attacks on DRL-based policies is essential. Such research not only reveals the vulnerabilities of current approaches but also, more importantly, provides valuable insights for designing more robust defense algorithms. Ultimately, it paves the way for trustworthy and reliable autonomous driving technologies.

A. Motivation

Existing studies have demonstrated the high vulnerability of DRL-based autonomous driving policies to adversarial attacks [14], [15], [16]. For instance, an attacker can manipulate the behaviors of autonomous driving agents by injecting adversarial perturbations into their observations, such as camera feeds and radar signals. Carefully designed perturbations can severely compromise the decision-making process of agents. Such disruptions may induce abnormal driving behaviors like sudden acceleration or abrupt braking, increasing the risk of accidents.

While existing studies on adversarial attacks have achieved notable results, they still face the following key challenges:

- 1) *Not Worst-Case Oriented*: Current studies typically operate under a simplified assumption that adversaries aim solely to minimize agent rewards or maximize policy deviation [17], [18], [19], [20], [21]. Moreover, existing research primarily associates the agent's rewards with driving efficiency. As a result, such attacks typically lead to relatively limited consequences, such as reduced ride comfort and traffic efficiency. While these attacks effectively achieve their intended objectives, they fail to

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induce the agent into worst-case scenarios (e.g., collision). In other words, optimizing autonomous driving policies under these methods may fail to guarantee safety.

- 2) *Not Stealthy*: Most studies adopt an overly idealized assumption: adversaries can inject adversarial perturbations at every time step [17], [18], [19], [20], [21], [22], [23], [24]. However, such high-frequency attack patterns are easily detectable and fail to reflect genuine threats. More critically, such high-frequency attack patterns may exceed the normal operational boundaries of autonomous driving models, thereby making it difficult to accurately identify the vulnerability of policies to perturbations.
- 3) *Not Efficient*: Current high-frequency attack methods also have the issue of efficiency. These methods [17], [18], [19], [20], [21], [22], [23], [24] typically employ gradient-based techniques for adversarial perturbation generation, which requires multiple iterations. As a result, high-frequency attacks introduce significant computational overhead in both the training and deployment phases. This ultimately affects the overall efficiency of the attack method.

B. Contribution

The above discussions motivate our work in this paper. We propose a novel adversarial attack method against DRL-based autonomous driving agents. Our method is specifically designed to induce safety violations with a minimal number of attack steps, achieving a high degree of stealth and efficiency. The attack problem is formulated as a mixed-integer optimization problem. Due to the difficulty of solving this problem, our method employs an adversary trained using the proximal policy optimization algorithm (PPO) [25] to solve the problem and obtain the optimal policy. Since the adversary's decisions are influenced by attack-related information, such as the origin actions of the agent, we incorporate this information into the adversary's observations to provide more decision-making context. Moreover, a trajectory clipping technique is adopted for data pruning to enhance sample quality. Our best knowledge indicates that we are the first to demonstrate such precise and stealthy adversarial attacks in autonomous driving scenarios, paving the way for further advancements in the robustness of autonomous driving policies. Specifically, compared to previous methods, our attack method provides the following key advantages.

- 1) *More Targeted*: Our attack method is driven by safety violations and aims to induce the agent into worst-case scenarios through targeted attacks. In other words, our method can better uncover the vulnerabilities of autonomous driving policies, thereby promoting the development of trustworthy autonomous driving.
- 2) *Significantly Reduced Attack Frequency*: In our method, we introduce an upper limit on the number of attacks to constrain the adversary. Thus, the adversary must strategically launch attacks at critical moments to cause safety violations, rather than attacking at every time

step. This design reduces attack frequency while further ensuring the attack remains stealthy.

- 3) *Domain-Agnostic and Self-Learning*: In our attack method, the adversary is DRL-based. It autonomously identifies critical attack moments and launches attacks through continuous interaction with the environment and the driving agent. The entire process is self-learning and domain-agnostic. Moreover, the nature of DRL also enables the adversary to consider the long-term impact of attacks, further enhancing attack performance. The above features enable our method to adapt well to the complexity and diversity of autonomous driving scenarios.

The capabilities of our attack method are validated through the experiments in two critical scenarios: 1) unprotected left turn and 2) on-ramp merging. The simulations are conducted based on SUMO [26]. The results show that our method can achieve collision rates (CRs) of over 90%, with an average of only three attack steps in most cases. Moreover, compared to other baseline attack methods, our method enhances the attack efficiency (AE) by at least 130% in all cases.

The remaining article is structured as follows. Section II introduces the related work. Section III outlines the preliminaries on Markov decision processes (MDPs) and formulates the attack problem. In Section IV, we detail our proposed attack method. The experimental results are discussed and analyzed in Section V, followed by the conclusion in Section VI.

II. RELATED WORK

This section first reviews the recent research on DRL-based autonomous driving policies in Section II-A, followed by a review of adversarial attack methods targeting these policies in Section II-B. Section II-C discusses existing stealthy and efficient adversarial attack methods for general tasks. Section II-D summarizes the gaps between our work and existing approaches.

A. DRL-Based Autonomous Driving

The rapid advancements in machine intelligence and data-driven approaches have propelled DRL to the forefront of autonomous driving policies [27]. For instance, Huang et al. [28] applied human prior knowledge and DRL to tackle the challenges of sample efficiency and reward function design, which is verified in unprotected left-turn and roundabout scenarios. Li et al. [29] proposed a hierarchical skill-based offline reinforcement learning method to solve the long-horizon vehicle planning task. Dang et al. [30] developed an event-triggered model predictive control framework based on DRL to address the path following problem in autonomous driving. Although the above DRL-based driving policies have achieved remarkable advancements in many scenarios, the lack of robustness against adversarial attacks limits their application in real-world settings [31].

B. Attack Methods for DRL-Based Autonomous Driving

Existing studies have demonstrated that well-trained DRL policies remain highly vulnerable when facing adversarial

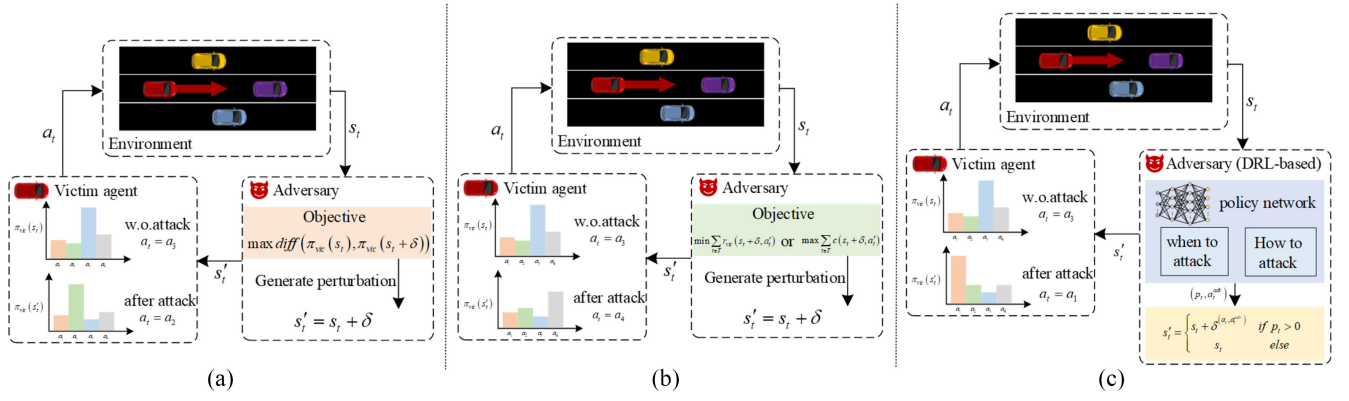


Fig. 1. Difference between the existing attack paradigms and our attack method. (a) Untargeted attack [28], [29], [30], [33]. (b) Targeted [14], [31], [32], [34]. (c) Ours.

perturbations [32], [33], [34], [35]. Adding adversarial perturbations to state inputs in autonomous driving policies can cause sudden acceleration or braking of the ego vehicle, posing a safety risk. Consequently, some studies attempted to explore the robustness of autonomous driving policies under adversarial environments and have proposed some adversarial attack methods. These attack methods can be broadly categorized into two types: 1) untargeted attacks [18], [19], [20] and 2) targeted attacks [15], [22], [23].

Untargeted attacks typically aim to maximize the distance between the agent's policies before and after the perturbation, as shown in Fig. 1(a). He et al. [21] proposed a black-box attack technique with Bayesian optimization to maximize the action deviation of the ego vehicle by introducing adversarial perturbations to its observations. However, these methods fail to induce worst-case scenarios, such as collisions, due to the lack of a clear purpose.

In contrast, targeted attack methods are designed to achieve specific harmful outcomes, such as minimizing the agent's reward or maximizing safety violations, as shown in Fig. 1(b). For instance, Ma et al. [24] introduced an adversary that can introduce up to 20% adversarial perturbations to the actions of the autonomous driving agent to minimize the agent's reward.

All the above methods assume that adversarial perturbations are continuously applied throughout the driving process. However, excessive perturbations make the attack more detectable and easier to mitigate. Furthermore, continuous attacks fail to significantly improve the effectiveness, as the critical moments for attacks are sparse. Noncritical perturbation samples may also interfere with the adversarial training of the autonomous driving agent, hindering its ability to improve robustness. Thus, it is essential to develop stealthy and efficient attack methods which launch attacks at critical moments.

C. Stealthy and Efficient Adversarial Attack Methods for DRL

Some explorations have been conducted in the field of DRL to improve the efficiency and stealth of attacks. Lin et al. [32] proposed a strategically-timed attack method that triggers an attack when the value of the relative action preference function (RAPF) exceeds a precalculated threshold. Li et al. [33] further considered the differences in the probabilities of all actions

based on RAPF, making the judgment of critical moments more precise. Sun et al. [34] proposed a critical point attack in which the adversary constructs a domain-specific model to predict subsequent states and assess the damage of each potential attack policy. A damage awareness metric is used as a threshold to determine critical moments.

Although the above methods have achieved promising results, their reliance on manually designed domain-specific metrics limits their effectiveness and scalability in more complex scenarios. As a result, some studies have attempted to leverage the potential of DRL to create an adversary capable of both selecting critical moments and executing attack actions. Lin et al. [32] proposed an antagonist attack in which a DRL-based adversary is trained to learn both when-to-attack and how-to-attack policies, guided by the reward function of the victim agent. However, this approach overlooks the sparsity of perturbation injection, which may lead to suboptimal attack policies. Mo et al. [35] proposed a decoupled adversarial policy in which trajectory clipping and trajectory padding are used to mitigate the impact of sparse perturbation injection. Moreover, a preconstructed database is used to provide universal perturbations to induce the victim agent to choose adversarial actions with 100% probability. However, designing universal perturbations for autonomous driving tasks with continuous action space and complex state space is challenging.

D. Research Gaps

Compared to existing works, our research addresses the following critical gaps.

- 1) Different from existing untargeted attacks [18], [19], [20], [21], we focus on maximizing safety violations of autonomous driving agents, making our attack more targeted and practically valuable.
- 2) Different from existing high-frequency attack methods [18], [19], [20], [21], [22], [23], [24], our approach considers stealth. Moreover, launching attacks only at critical moments enhances both efficiency and effectiveness.
- 3) Different from existing stealthy adversarial attack methods [32], [34], [35], which are primarily designed for general tasks, our method specifically targets the

unique challenges of autonomous driving scenarios, such as complex state and action space.

Building on these insights, we propose a novel stealthy and efficient adversarial attack method against DRL-based autonomous driving agents, as shown in Fig. 1(c).

III. PROBLEM FORMULATION

This section first describes the preliminary concept of MDP in Section III-A. Section III-B outlines the assumptions regarding the adversary's knowledge and capabilities. Then we model the attack problem as a mathematical formulation according to the goal of the proposed attack in Section III-C.

A. Preliminary

Driving policies can be learned through DRL algorithms [1]. Before learning the policy, it is usually necessary to model the driving policy learning problem as an MDP which can be defined by a tuple $(\mathcal{S}, \mathcal{A}, r, \mathcal{P}, \gamma)$. Here state space $\mathcal{S}$ denotes the set of all possible states of the system. In autonomous driving, the state information typically includes information about the ego vehicle, surrounding vehicles, and driving environment. Action space $\mathcal{A}$ represents the set of all possible control actions, such as steering and throttle. The reward function $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is designed to guide the agent to learn the desired driving policy and $\mathcal{P} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ describes the transition probability of the system. $\gamma \in (0, 1)$ is the discount factor. The DRL-based agent aims to find an optimal policy π^* that maximizes the cumulative discounted reward

$$J(\pi) = \mathbb{E}_{a_t \sim \pi(s_t)} \left(\sum_{t=1}^T \gamma^t r(s_t, a_t) \right). \quad (1)$$

B. Assumptions

Before formulating the attack problem, we first define the following assumptions regarding the adversary's knowledge and capabilities:

Knowledge of the Adversary: The adversary can access the observations and actions of the autonomous driving agent by compromising the autopilot electronic control unit [36], [37]. Moreover, the adversary requires knowledge of the autonomous driving model to generate adversarial perturbations. We consider this assumption is reasonable since the attacker could extract the autonomous driving model by reverse-engineering the deployed autonomous system [38], [39].

Capabilities of the Adversary: To ensure the stealth of the attack, the adversary is only permitted to modify the agent's observations by injecting adversarial perturbations [40]. This can be realized through CAN bus injection techniques [41], [42].

C. Problem Statement

In this article, we aim to find a stealthy and efficient attack method against DRL-based autonomous driving policies. Compared to launching attacks at every time step, a

more efficient and stealthy strategy is to launch fewer attacks while still achieving the desired objectives. Let $\bar{r}$ as the reward function of the adversary, the details of which are defined in Section V-B. We thus formulate the attack as the following optimization problem:

$$\max_{(\mathbf{x}, \mathbf{a}_{\text{adv}})} \mathbb{E}_{a''_t \sim \pi(s'_t)} \left(\sum_{t=1}^T \gamma^t \bar{r}(s'_t, a''_t) \right) \quad (2)$$

$$\text{s.t. } \delta_t = \text{PG}(\pi, s_t, a'_t), t \in T \quad (2a)$$

$$s'_t = s_t + x_t \delta_t, t \in T \quad (2b)$$

$$\delta_t \leq \epsilon, t \in T \quad (2c)$$

$$\sum_{t \in T} x_t \leq \Gamma, \Gamma \ll T. \quad (2d)$$

Equation (2) is the objective function that maximizes the adversary's cumulative discounted reward over the entire episode, where T is the maximum length of each episode. The purpose of adopting cumulative discounted rewards is to implicitly drive the adversary to complete the attack in as few time steps as possible. The variables are denoted as $(\mathbf{x}, \mathbf{a}_{\text{adv}})$, where $\mathbf{x} = \{x_t, t \in T\}$ is a set of binary variables used to control whether to apply the attack at each time step, and $\mathbf{a}_{\text{adv}} = \{a'_t, t \in T\}$ denotes the set of actions that the adversary expects the victim to perform at each time step. If $x_t = 1$, the adversary will invoke the perturbation generation function (PG) to generate perturbation δ_t , as shown in (2a). The inputs of PG includes the agent's policy π , the current state s_t , and a'_t . Eq. (2b) illustrates the perturbation injection process. When the adversary decides to launch an attack ($x_t = 1$), it adds δ_t to the current state s_t , forming a new state s'_t . Subsequently, the agent will generate and execute the action a''_t based on s'_t . To reduce the probability of the attack being detected, we set a threshold ϵ for δ_t , as shown in (2c). To control the number of attacks launched by the adversary, we introduce an attack upper limit Γ , as illustrated in (2d). Solving this optimization problem is a challenge because it is a complex mixed integer optimization problem. The difficulty is further complicated by the nonlinear growth in the size of the problem, driven by T [32].

IV. METHODOLOGY

This section first outlines the overall framework of our attack method in Section IV-A, followed by a detailed description of the algorithm implementation and the trajectory clipping technique in Sections IV-B and Section IV-C, respectively.

A. Framework

The optimization problem (2) can be easily divided into two subproblems: 1) *when-to-attack* and 2) *how-to-attack*. Our attack method introduces a DRL-based adversary to independently learn the optimal policies for both subproblems. As illustrated in Fig. 2, the method operates in two stages: decision-making and attack launching. In the decision-making stage, solutions to the two subproblems are generated, while

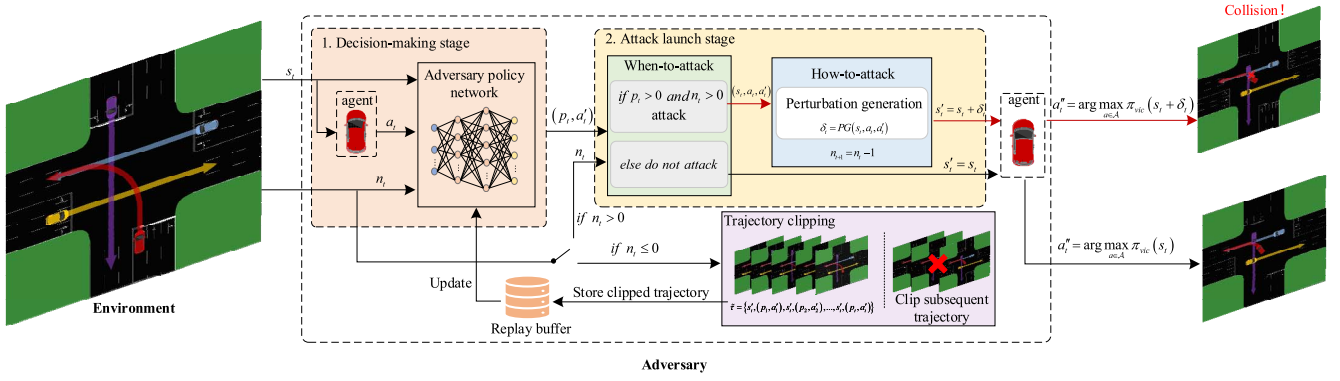


Fig. 2. Framework of our attack method.

the attack launching stage executes attacks based on these solutions. The details of each stage are as follows.

Decision-Making Stage: Where the adversary needs to make attack decisions $(p_t, a'_t) \sim \pi_{\text{adv}}(s_t^{\text{adv}})$ based on its observation s_t^{adv} . Besides treating $s_t^{\text{adv}} = s_t$ as in [34], we introduce additional attack-related information in order to enhance the ability of the adversary. It is achieved by adding the remaining number of attacks n_t and the agent's original action a_t into the adversary's observation, i.e., $s_t^{\text{adv}} = (s_t, n_t, a_t)$. p_t and a' respectively represent the solutions to the subproblems. Specifically, $p_t \in [-1, 1]$ determines whether to inject the perturbation into s_t to launch an attack, defining the *when-to-attack* solution. $a'_t \in \mathcal{A}$ represents the lure action that the adversary aims to induce the agent to take through adversarial perturbations, defining the *how-to-attack* solution.

Attack Launching Stage: In this stage, the adversary will generate the perturbation δ_t to mislead the agent to trigger the action a'_t , if $p_t \geq 0$. Conversely, if $p_t < 0$, it indicates that no attack will be launched at time step t . After launching an attack, the adversary will terminate the episode if $n_t = 0$. This process, referred to as trajectory clipping, aims to balance the sample distribution and improve the training efficiency of the adversary.

Our attack method is capable of learning optimal *when-to-attack* and *how-to-attack* policies automatically without any domain knowledge. The domain-agnostic and self-learning characteristics make our method highly suitable for complex and diverse autonomous driving scenarios. Moreover, the introduction of additional attack-related information and trajectory clipping further enhances the effectiveness of our attack method.

B. Algorithm

Given that our autonomous driving scenario involves a continuous action space, we choose PPO, a state-of-the-art on-policy DRL algorithm [43], to train the adversary. PPO consists of two networks, an actor network $\pi_{\theta}^{\text{adv}}$ and a critic network V_{ϕ}^{adv} , where θ and ϕ are the parameters corresponding to these two networks. Given an adversary observation s_t^{adv} , $\pi_{\theta}^{\text{adv}}$ will output the policy $\pi_{\theta}^{\text{adv}}((p_t, a'_t)|s_t^{\text{adv}})$. and V_{ϕ}^{adv} will estimate the value of s_t^{adv} , which can be expressed as

$$V_{\phi}^{\text{adv}}(s_t^{\text{adv}}) = \mathbb{E} \left[\sum_{k=1}^{T-t} \gamma^{k-1} \bar{r}_{t+k} | s_t^{\text{adv}} \right]. \quad (3)$$

The loss function used to optimize $\pi_{\theta}^{\text{adv}}$ can be expressed as

$$L_a(\theta) = \mathbb{E}_t \left[\min \left(\rho_t^{\text{clip}}(\theta), \rho_t(\theta) \right) \hat{A}_t \right] \quad (4)$$

where

$$\rho_t^{\text{clip}}(\theta) = \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \quad (5)$$

$$\rho_t(\theta) = \frac{\pi_{\theta}^{\text{adv}}((p_t, a'_t)|s_t^{\text{adv}})}{\pi_{\theta}^{\text{adv,old}}((p_t, a'_t)|s_t^{\text{adv}})} \quad (6)$$

where $\rho_t(\theta)$ represents the probability ratio between the current policy $\pi_{\theta}^{\text{adv}}$ and the old policy $\pi_{\theta}^{\text{adv,old}}$. $\hat{A}_t$ is the estimated advantage and can be defined as

$$\hat{A}_t = \sum_{i=t}^{T-1} (\gamma \lambda)^{i-t} \left(\bar{r}_i + \gamma V_{\phi}^{\text{adv}}(s_{i+1}^{\text{adv}}) - \gamma V_{\phi}^{\text{adv}}(s_t^{\text{adv}}) \right) \quad (7)$$

where λ is a tuning parameter.

The loss function for the critic network is typically a mean squared error, which can be expressed as

$$L_c(\phi) = \mathbb{E}_t \left[\left(V_{\phi}^{\text{adv}}(s_t^{\text{adv}}) - \left(\bar{r}_t + \gamma V_{\phi}^{\text{adv}}(s_{t+1}^{\text{adv}}) \right) \right)^2 \right]. \quad (8)$$

Then we can obtain the total loss function as

$$L(\theta, \phi) = L_a(\theta) - c_1 L_c(\phi) + c_2 S[\pi_{\theta}^{\text{adv}}](s_t^{\text{adv}}) \quad (9)$$

where c_1 and c_2 are coefficients and $S[\cdot]$ denotes an entropy bonus. Based on the loss function, θ and ϕ can be simultaneously updated using gradient-based methods.

A step-by-step outline of our attack method is given in Algorithm 1. First, initialize the network parameters θ and ϕ , as well as the experience replay buffer D . Next, the adversary will interact with the environment for L episodes (line 1), with each episode having a maximum length of T (line 3). At the beginning of each episode, initialize $n_t = \Gamma$ (line 2). At each time step, the adversary generates (p_t, a'_t) based on its observations (line 4). Then the adversary determines whether to add perturbations to the agent's observations based on p_t and n_t (lines 5–10). After that, the agent will generate an action based on s'_t and execute it (lines 11 and 12). The returned next observation s_{t+1} and reward $\bar{r}_t$ will be stored into D along with s_t and (p_t, a'_t) (line 13). Finally, the adversary determines whether to perform trajectory clipping based on n_t (line 15) and samples data from the buffer and updates the policy parameters according to (9) (line 17).

Algorithm 1 Our Attack Method

Input: Environment $\mathcal{E}$, victim agent policy π_{vic}
Output: The optimal adversarial policy network π_{adv}^*
Initialize network parameters θ and ϕ , experience replay buffer D

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1: for episode = 1, 2, ...,  $L$  do:
2:    $n_1 = \Gamma$ 
3:   for  $t = 1, 2, \dots, T$  do:
4:      $a_t = \arg \max_{a \in \mathcal{A}} \pi_{vic}(a|s_t), (p_t, a'_t) \sim \pi_{adv}(s_t, a_t, n_t)$ 
5:     If  $p_t > 0$  and  $n_{t-1} > 0$ 
6:        $n_{t+1} = n_t - 1$ 
7:        $\delta = PerturbationGenerate(s_t, a_t, a'_t)$ 
8:        $s'_t = s_t + \delta$ 
9:     ELSE
10:       $s'_t = s_t, n_{t+1} = n_t$ 
11:       $a''_t = \pi_{vic}(s'_t)$ 
12:      Perform  $a''_t$  and obtain  $(s_{t+1}, \bar{r}_t)$ 
13:      Store  $\tau_t = (s_t, (p_t, a'_t), s_{t+1}, \bar{r}_t)$  into  $D$ 
14:      If  $n_{t+1} = 0$ 
15:        Clip the trajectory and reset the environment  $\mathcal{E}$ 
16:      END for
17:      Sample from  $D$  to update  $\theta$  and  $\phi$  according to (9)
18:    END for

```

C. Trajectory Clipping

Since we limit the number of attack attempts per episode to ensure efficiency, once the number of launched attacks reaches the maximum number Γ , the adversary will be unable to execute any further actions in subsequent steps, even if $p \geq 0$. In this case, continuing the episode and storing the subsequent trajectory in the buffer could be harmful. This is because storing invalid samples (subsequent trajectories) in the buffer would lead to data imbalance, thereby impacting the adversary's ability to learn an effective policy. Furthermore, the agent's policy may trigger safety violations in subsequent time steps without adversarial perturbations. These samples will generate positive rewards to the adversary. However, since these safety violations are not caused by the adversary, adding these samples in the buffer will mislead the adversary's policy learning.

We believe that an ideal attack should immediately cause the safety violations after the attack is launched. It is possible that the safety violation may occur several time steps after the attack is launched, but it is difficult to determine whether this situation is caused by the adversary or the agent. Therefore, we adopt a trajectory clipping technology to terminate the current episode directly when $n_t = 0$. Specifically, assuming t' as the time step when $n_{t'} = 0$. There are two possible scenarios at this point: one is that the attack causes a collision, in which case the episode ends naturally and a complete trajectory τ is obtained. The other is that no collision is triggered. Normally, the agent will continue driving until time step T . However, as mentioned above, the trajectory $\bar{\tau}$ after t' is harmful and thus needs to be clipped. We define the trajectory $\tilde{\tau}$ until t' and $\bar{\tau}$ as follows:

$$\tilde{\tau} = \{s_1^{\text{adv}}, (p_1, a'_1), s_2^{\text{adv}}, (p_2, a'_2), \dots, s_{t'}^{\text{adv}}, (p_{t'}, a'_{t'})\} \quad (10)$$

$$\bar{\tau} = \{s_{t'+1}^{\text{adv}}, (p_{t'+1}, a'_{t'+1}), \dots, s_T^{\text{adv}}, (p_T, a'_T)\}. \quad (11)$$

After clipping, only the samples in $\tilde{\tau}$ are used for training. Since the trajectory is clipped, we need to redefine the rewards for these samples. For the vanilla policy training, the accumulated discounted rewards can be expressed as

$$R_{\text{adv}} = \mathbb{E}_{\substack{a_t \sim \pi_{vic}(s'_t), \\ (p_t, a'_t) \sim \pi_{adv}(s_t, n_t, a_t)}} \left[\sum_{t=1}^T \gamma^t \bar{r}(s_t, (p_t, a'_t)) \right]. \quad (12)$$

The accumulated discounted rewards after clipping should be [35]

$$R_{\text{adv}}^{\text{clip}} = \mathbb{E} \left[\sum_{t=1}^{t'-1} \gamma^t \bar{r}(s_t, (p_t, a'_t)) + \gamma^{t'} \sum_{t=t'}^T \bar{r}(s_t, (p_t, a'_t)) \right]. \quad (13)$$

Since $\bar{r}(s_t, (p_t, a'_t))$ is related to safety violations, and we assume that the safety violations after t' are independent of the adversary, we do not consider the reward after t' . Therefore, (13) can be simplified to

$$R_{\text{adv}}^{\text{clip}} = \mathbb{E} \left[\sum_{t=1}^{t'} \gamma^t \bar{r}(s_t, (p_t, a'_t)) \right]. \quad (14)$$

V. EXPERIMENT EVALUATION

We validate the proposed attack method on the SUMO platform. First, we introduce the relevant experimental settings in Sections V-A–V-E, including environment settings, the MDP, baselines, training details, and metrics. Then we organize our experiments and explain the results to address the following key questions.

- 1) *RQ1*: Can the proposed attack method can achieve the adversarial goal within a limited number of attacks? (Studied in Section V-F1)
- 2) *RQ2*: Does the proposed attack method outperform the baseline and state-of-the-art methods? (Studied in Section V-F2)
- 3) *RQ3*: How does the value of Γ affect the proposed attack method? (Studied in Section V-F3)
- 4) *RQ4*: How does the performance of our proposed attack method vary under different traffic densities? (Studied in Section V-F4)
- 5) *RQ5*: How does each component contribute to the proposed attack method? (Studied in Section V-G)

A. Environment Setting

We evaluate the performance of our method in two critical scenarios: unprotected left-turn and on-ramp merging, as shown in Fig. 3. In the simulated environment, all vehicles employ the LC2013 model of SUMO for lane-changing behavior. To ensure realistic dynamics, the maximum speed v_{max} and maximum acceleration (and deceleration) β of the ego vehicle are set to 15 m/s and 7.6 m/s², respectively. The traffic flow density is determined by the probability p of vehicle arrivals per second. If there is no special explanation, we set $p = 0.5$.

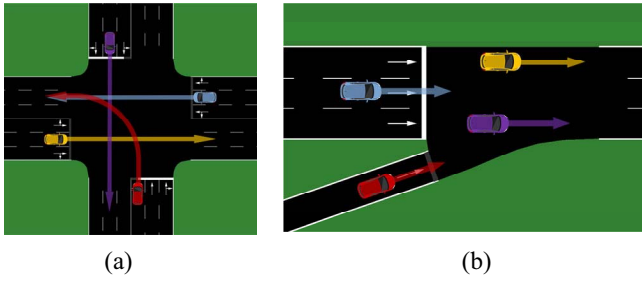


Fig. 3. Experimental scenarios. (a) Unprotected left-turn. (b) On-ramp merging.

B. Markov Decision Process

Based on the simulation environment, we construct an MDP with the following details:

State Space $\mathcal{S}$: The state $s_t \in \mathcal{S}$ consists of two components. One is the speed and velocity direction of the ego vehicle. The other part contains information about the six nearest vehicles within a 200-m radius of the ego vehicle, including the relative distance, orientation, speed, and velocity direction. The positions of these six cars are front, rear, left front, left rear, right front and right rear.

Action Space $\mathcal{A}$: At time step t , the action of the agent a_t involves continuous acceleration or deceleration, represented as a continuous variable within the range $[-1, 1]$. To accurately reflect real conditions, a_t needs to be multiplied by the acceleration coefficient β to obtain the actual acceleration of the ego vehicle. Thus, the velocity update equation can be obtained as $v_{\text{ego}}(t+1) = v_{\text{ego}}(t) + a_t \beta \Delta t$.

Reward Function r : The reward function of the victim agent r_t consists of two components: 1) efficiency reward and 2) safety penalty, which can be expressed as

$$r_t = \frac{v_t}{v_{\max}} - c(s_t, a''_t)$$

$$c(s_t, a''_t) = \begin{cases} 1, & \text{if collision} \\ 0, & \text{else} \end{cases} \quad (15)$$

where $v_{\max}$ denotes the maximum speed of the ego vehicle and a''_t represents the action performed by the ego vehicle at time step t . Specifically, the goal of the victim agent is to drive the ego vehicle as quickly as possible while avoiding collisions. Unlike existing attack methods that simply set the adversary's reward as $\bar{r}_t = -r_t$ [35], we propose a worst-case oriented reward function. Specifically, we argue that the adversary should focus on inducing safety violations, such as collisions. Therefore, we adopt a binary reward function based on collision events, where $\bar{r} = c(s_t, a''_t)$. The adversary receives a reward of 1 when successfully inducing a collision, and 0 otherwise. This concise design also improves cross-scenario adaptability of our method, as $\bar{r}$ depends solely on a single factor and remains unaffected by other variables.

C. Baselines

We select autonomous driving agents based on state-of-the-art RL algorithms to verify the effectiveness of our attack method. These agents can be divided into three categories:

Vanilla RL: The baselines include PPO and the state-of-the-art off-policy algorithms SAC [44] and TD3 [45].

Safe RL: This category of methods avoids the agent selecting dangerous actions by introducing constraint mechanisms. We adopt FNI-RL [46], a state-of-the-art method in this category, as our baseline approach.

Robust RL: This category of methods enhances the robustness of algorithms by incorporating adversarial perturbations. We selected two algorithms: the first is DARRL [17], which improves the robustness of the agent by considering the cost induced by adversarial perturbations in the optimization objective. The second is adversarial RL (ARL), a widely-used approach that enhances policy robustness by injecting perturbations directly into the training process of the agent. In our setup, ARL injects random adversarial perturbations at each time step with a probability of 20%.

In terms of attack methods, we selected the following attack methods for comparison with our proposed attack method.

Random Attack (RA): At each time step, the adversary randomly samples an adversarial action from the action space. Then perturbations are generated to guide the agent toward selecting an action aligned with the adversarial one. Since adversarial actions are randomly selected, RA is a typical nontargeted attack method.

Action Modification Attack (AMA): In this attack, the adversary possesses nearly the same capabilities as our attack method. The difference is that this attack directly forces the adversarial action as the final executed action, rather than guiding the agent through perturbations, i.e., $a''_t = a'_t$. It is typically unrealistic to directly manipulate the agent's action in practical applications, as it requires the adversary to have higher privileges. However, this attack method can serve as a strong baseline to verify the effectiveness of our attack. Specifically, the effect of this attack is intuitively not weaker than our attack method since each attack can achieve the expected target with a 100% success rate (SR).

Unlimited Attack (UA): This attack is similar to the attack methods commonly adopted in current work, as it performs an attack at each time step. This attack generates the adversarial action in the same way as our method.

D. Training Details

To verify the effectiveness of our attack method, we first train several victim agents using different DRL algorithms. All agents were trained for 12 000 steps, with a maximum episode length T set to 30. All DRL algorithms are implemented using Stable Baselines3 [47]. We fine-tune the hyperparameters of the PPO-based victim agent for better performance. Specifically, the batch size is set to 256, the learning rate to 0.001, and the number of epochs to 15. The parameters for the other agents are set to the default values provided by Stable Baselines3. With the exception of the RA baseline, all other attack methods utilize PPO as their underlying algorithm. It is worth emphasizing that our proposed method is not algorithm-specific, and the adoption of PPO is primarily motivated by its dominant presence in contemporary reinforcement learning

applications. We train the adversaries based on the victim agents to generate the corresponding attack policies. The total training steps and maximum episode length are consistent with the victim agents, and the hyperparameters, such as learning rate and batch size are set to default values.

In terms of perturbation generation methodology, we select two widely-adopted approaches: basic iterative method (BIM) [20], [48] and projected gradient descent (PGD) [49]. BIM generates adversarial examples through iterative gradient ascent with controlled step sizes α while keeping the perturbation within a given bound ϵ . PGD extends BIM by incorporating random initialization and projection to maintain perturbation validity. To ensure fair comparison, both methods were configured with identical parameter settings: perturbation budget $\epsilon = 0.1$ and iteration number $T_{\text{iter}} = 0.1$. Furthermore, for BIM implementation, we set the step size $\alpha = 0.02$.

All the experiments are conducted on a server equipped with Intel Xeon Gold 6230 CPUs and NVIDIA GeForce RTX 4090 GPUs.

E. Metrics

We select multiple metrics to conduct a comprehensive evaluation of our attack method. Besides the common metrics such as SR, CR, average speed (AS), and average reward (AR) [46], we also adopt the average number of attacks (ANA) to evaluate the efficiency of our attack method. Specifically, ANA refers to the ANA launched by the adversary in an episode. Moreover, we introduce a new metric to evaluate the effectiveness of a single attack, called AE. AE can be calculated using CR and ANA, as follows:

$$AE = CR \cdot e^{-k \cdot \text{ANA}} \quad (16)$$

where k is a parameter used to adjust the weights of CR and ANA. In this article, it is set to 0.05. All metrics are computed based on the results of 100 test episodes.

F. Performance Evaluation

In this section, we aim to address RQ1-RQ4 through experiments and rigorous analysis.

1) *Effectiveness of Our Method*: In this section, we aim to verify that our method enables stealthy and efficient attacks, maximizing the CR with a limited number of attacks. We evaluate the performance of our method against victim agents in the unprotected left turn scenario. The results are demonstrated in Fig. 4. It can be observed that, in most cases, our attack method achieves a CR of around 90% within an average of three attacks per episode. Both BIM-based and PGD-based attacks achieve similar attack effectiveness, i.e., CR.

2) *Performance Comparison*: To further evaluate the effectiveness of our method, we compare the performance of our method with the baseline attack methods. Table I presents the experimental results for the unprotected left-turn scenario. Compared to AMA and UA, our method can achieve optimal or near-optimal performance in CR. In contrast to our method and AMA, the effectiveness of UA relies on

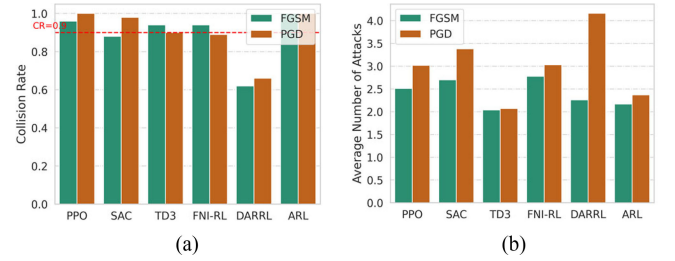


Fig. 4. Performance of our attack method against different victim agents. (a) CR. (b) ANA.

the continuous addition of perturbation. Such high-frequency attack patterns not only significantly increase detection risks but also substantially reduce training efficiency. Experimental results demonstrate that the UA method typically requires more than twice the training time compared to our method. Detailed analysis of the training efficiency of our method is presented in the supplementary file.

Leveraging the precision of action modification, AMA achieves a 100% CR in most cases. However, AMA is generally more detectable than UA due to directly modifying actions. Nevertheless, we select AMA as a baseline to serve as an ideal case for our attack method. Although direct action modification enables AMA to perform well, its deterministic nature may restrict the adversary's exploration of the state space. In RL, insufficient exploration may lead the agent to get trapped in local optima. As a result, in certain cases, AMA requires more attack steps to achieve a CR comparable to that of our method. Finally, we compared our attack method with other algorithms to validate its AE. Compared to UA and RA, our method achieves an average improvement of 115.81% and 547.90% in AE, respectively. The results and corresponding analysis for the on-ramp merging scenario can be found in the supplementary file.

3) *Impact of the Upper Bound of Attack Steps*: In this section, we aim to explore the impact of the upper limit Γ on ANA. Fig. 5 shows the performance of our attack method under different Γ . The victim agent is based on FNI-RL, and the perturbation generation method is BIM. The vertical axis denotes the adversary's binary collision reward. We employ convolutional smoothing to facilitate visualization, making the vertical axis an approximation of CR. It can be observed that as Γ increases, the CR increases, indicating an improvement in the attack capability of the adversary. This is because a larger upper bound enables the adversary to explore more situations, thereby increasing the likelihood of learning the optimal attack policy. Similar conclusions can be drawn in other cases where the victim agent and perturbation generation method are different.

However, increasing Γ does not necessarily yield increased ANA. To prove this, we test the attack policies obtained under different Γ with 100 episodes, and the results are shown in Table II. The experimental results indicate that the ANA for all policies eventually converges to around 3, showing a significant gap compared to Γ . This clearly demonstrates that increasing Γ does not result in a higher ANA. During

TABLE I
STATISTICAL RESULTS OF AUTONOMOUS DRIVING AGENTS UNDER DIFFERENT ADVERSARIAL ATTACK METHODS (UNPROTECTED LEFT TURN)

Method	Metrics	No Attack	AMA	RA		UA		Ours	
				BIM	PGD	BIM	PGD	BIM	PGD
PPO	SR	0.90	0.00	0.47	0.61	0.02	0.05	0.04	0.04
	CR	0.10	1.00	0.46	0.39	0.98	0.95	0.96	0.96
	ANA	-	1.59±1.19	25.77±4.94	25.62±5.64	17.42±3.69	18.40±4.41	2.51±1.15	2.66±1.56
	AE	-	0.92	0.13	0.11	0.41	0.38	0.85	0.84
SAC	SR	1.00	0.00	0.52	0.60	0.06	0.21	0.12	0.02
	CR	0.00	1.00	0.47	0.39	0.94	0.79	0.88	0.98
	ANA	-	3.90±1.92	24.58±6.15	25.75±5.60	20.12±4.26	22.78±5.79	2.70±0.83	3.38±2.03
	AE	-	0.82	0.14	0.11	0.34	0.25	0.77	0.83
TD3	SR	0.72	0.06	0.19	0.24	0.00	0.01	0.06	0.10
	CR	0.20	0.94	0.76	0.73	1.00	0.99	0.94	0.90
	ANA	-	1.92±0.39	22.21±5.72	22.93±5.73	17.42±4.10	17.83±4.09	2.04±0.53	2.07±0.75
	AE	-	0.85	0.25	0.23	0.42	0.41	0.85	0.81
FNI-RL	SR	0.98	0.06	0.45	0.60	0.00	0.03	0.06	0.11
	CR	0.02	0.94	0.49	0.38	1.00	0.97	0.94	0.89
	ANA	-	2.82±2.02	23.89±6.42	26.00±5.45	16.72±2.80	17.06±3.66	2.78±2.22	3.03±2.15
	AE	-	0.82	0.15	0.10	0.43	0.41	0.82	0.76
DARRL	SR	0.98	0.04	0.53	0.67	0.12	0.20	0.38	0.34
	CR	0.02	0.96	0.41	0.30	0.88	0.80	0.62	0.66
	ANA	-	3.10±2.46	25.43±5.91	26.52±5.60	18.84±4.40	19.60±5.44	2.26±1.05	4.16±1.29
	AE	-	0.82	0.11	0.08	0.34	0.30	0.55	0.54
ARL	SR	0.84	0.00	0.67	0.64	0.00	0.12	0.00	0.00
	CR	0.14	1.00	0.32	0.36	1.00	0.88	1.00	1.00
	ANA	-	2.20±1.29	26.11±5.86	25.47±6.17	16.72±2.80	20.07±4.48	2.17±0.99	2.37±1.34
	AE	-	0.90	0.09	0.10	0.43	0.32	0.90	0.89

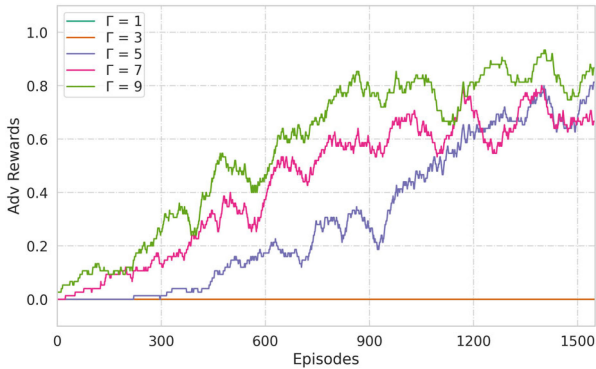


Fig. 5. Training reward curves under different upper bounds of attack steps.

the training process, the adversary learns the optimal attack policy, causing the number of attacks to eventually stabilize around a specific value. However, this does not imply that increasing Γ is futile. A higher Γ enables the adversary to conduct more comprehensive exploration. This accelerates attack policy training while moderately enhancing its final performance.

We then adjust the upper bound Γ_{test} during testing process to evaluate the performance of our method, as shown in Fig. 6. It can be observed that as Γ_{test} increases, the attack policies

TABLE II
CR, AND ANA OF ATTACK POLICIES UNDER DIFFERENT Γ

	CR	ANA		CR	ANA
$\Gamma = 5$	0.86	2.58±1.25	$\Gamma = 8$	0.88	3.27±2.30
$\Gamma = 6$	0.86	3.24±1.23	$\Gamma = 9$	0.94	3.10±2.05
$\Gamma = 7$	0.86	3.58±1.92	$\Gamma = 10$	0.94	3.00±2.30

trained under different Γ are able to achieve progressively higher CRs. The reason is that perturbation generation methods cannot guarantee a 100% SR. Therefore, it is necessary to launch subsequent attacks based on the outcome of previous attacks to mitigate the biases introduced by the perturbation generation method. Nevertheless, our method achieves more than 70% CR within three to four attack steps in most cases. This indicates that our attack method can achieve significant effectiveness by launching attacks in about 10% of the time steps within an episode, thus ensuring both stealth and efficiency.

4) *Performance Under Different Traffic Densities:* In this section, we aim to investigate the adaptability of pretrained attack policies across varying conditions. We construct two new unprotected left turn scenarios with different traffic densities, denoted as flow1 and flow2. Specifically, flow1

TABLE III
STATISTICAL RESULTS OF ATTACK METHODS UNDER DIFFERENT TRAFFIC DENSITIES

Method	PG	Metrics	PPO		SAC		TD3		FNI-RL		DARRL		ARL	
			Flow-1	Flow-2	Flow-1	Flow-2	Flow-1	Flow-2	Flow-1	Flow-2	Flow-1	Flow-2	Flow-1	Flow-2
No-attack	-	CR	0.32	0.06	0.12	0.06	0.20	0.16	0.02	0.00	0.14	0.00	0.14	0.04
		ANA	-	-	-	-	-	-	-	-	-	-	-	-
		AE	-	-	-	-	-	-	-	-	-	-	-	-
AMA	-	CR	0.78	0.92	0.76	0.88	0.84	0.98	0.70	0.96	0.80	1.00	0.68	0.78
		ANA	2.34	2.80	2.92	2.90	2.74	1.82	3.18	2.96	3.57	2.98	7.08	5.72
		AE	0.69	0.80	0.66	0.76	0.73	0.89	0.60	0.83	0.67	0.86	0.48	0.59
RA	BIM	CR	0.28	0.55	0.48	0.47	0.69	0.81	0.42	0.39	0.32	0.40	0.34	0.35
		ANA	27.66	25.16	24.61	25.02	22.98	22.20	25.33	25.27	26.06	25.37	27.73	27.80
		AE	0.07	0.16	0.14	0.13	0.22	0.27	0.12	0.11	0.09	0.11	0.08	0.09
	PGD	CR	0.35	0.42	0.43	0.51	0.60	0.75	0.38	0.42	0.39	0.32	0.37	0.32
		ANA	26.11	25.43	24.83	24.32	23.92	23.74	25.62	25.12	25.33	26.42	27.21	27.82
		AE	0.09	0.12	0.12	0.15	0.18	0.23	0.11	0.12	0.11	0.09	0.09	0.08
UA	BIM	CR	0.70	0.74	0.84	1.00	0.84	0.98	0.80	1.00	0.68	0.80	0.62	0.76
		ANA	22.66	22.08	18.98	17.50	19.40	17.30	19.70	16.74	20.76	19.88	25.24	25.52
		AE	0.25	0.41	0.33	0.41	0.32	0.41	0.30	0.43	0.24	0.30	0.18	0.21
	PGD	CR	0.85	1.00	0.84	0.97	0.84	0.97	0.78	0.97	0.70	0.72	0.61	0.51
		ANA	18.79	16.76	19.06	17.75	19.54	17.61	19.65	17.07	21.10	21.19	25.10	26.38
		AE	0.43	0.40	0.32	0.40	0.32	0.40	0.29	0.41	0.24	0.25	0.17	0.14
Ours	BIM	CR	0.78	0.92	0.74	0.88	0.82	0.92	0.64	0.96	0.54	0.44	0.60	0.88
		ANA	2.43	2.80	2.76	2.80	3.00	1.94	3.62	3.16	2.04	2.28	6.16	5.54
		AE	0.69	0.80	0.64	0.77	0.71	0.83	0.53	0.82	0.49	0.39	0.44	0.67
	PGD	CR	0.84	0.98	0.80	0.96	0.75	0.93	0.66	0.88	0.57	0.57	0.30	0.53
		ANA	3.12	3.32	3.51	3.38	2.08	1.84	3.67	2.84	3.33	2.19	1.67	6.57
		AE	0.72	0.83	0.67	0.81	0.68	0.85	0.55	0.76	0.48	0.51	0.28	0.38

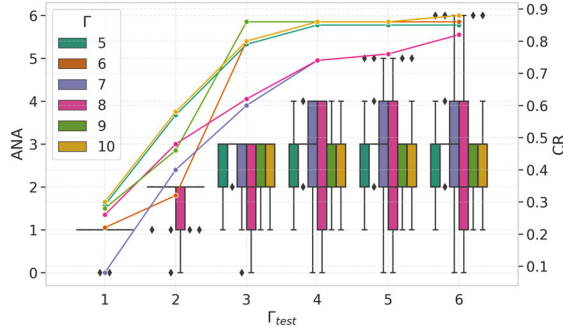


Fig. 6. Performance under different Γ_{test} .

and flow2 represent the cases where $p = 0.3$ and $p = 0.7$ respectively. Table III shows the performance of the attack methods under flow1 and flow2 scenarios. It can be observed that as traffic flow density increases, all attack methods can increase the CR in most cases. Similar to the results in Table I, our method consistently achieves comparable performance to AMA in all cases. This implies that existing perturbation generation methods can already yield satisfactory results. Our method outperforms both UA and RA across all scenarios, achieving comparable CRs with substantially reduced attack steps. Specifically, it demonstrates average AE improvements of 105.06% (versus UA) and 423.77% (versus RA) in flow1, and 109.03% (versus UA) and 443.37% (versus RA) in flow2.

G. Ablation Studies

We implement three ablation scenarios by removing different components of our attack method to evaluate the impact of the additional attack-related information and the trajectory clipping technology on the performance of our attack method. We select the case where the victim agent is FNI-RL and the perturbation generation method is BIM, the results are shown in Fig. 7. Similar conclusions can be drawn in other cases as well. “w.o.clipping” indicates that the trajectory is not clipped when $n_t = 0$. “w.o.v_action” denotes the exclusion of the victim agent’s original actions from the adversary’s observation space. Similarly, “w.o.remain_attack_times” represents the removal of the remaining attack steps information. It can be observed that both “w.o.v_action” and “w.o.remain_attack_times” maintain reward values below 0.4 throughout the training process, indicating that they have not learned effective attack policies. This failure highlights the critical role of additional attack-related information in enhancing the adversary’s decision-making process. Although the performance of “w.o.clipping” is relatively better, with its rewards increasing to 0.6 after 2000 episodes and still showing a trend of improvement, it suffers from a significant decline in training efficiency compared to our method. This highlights the crucial role of trajectory clipping in enhancing training efficiency by effectively filtering out invalid samples and allowing more focused policy learning.

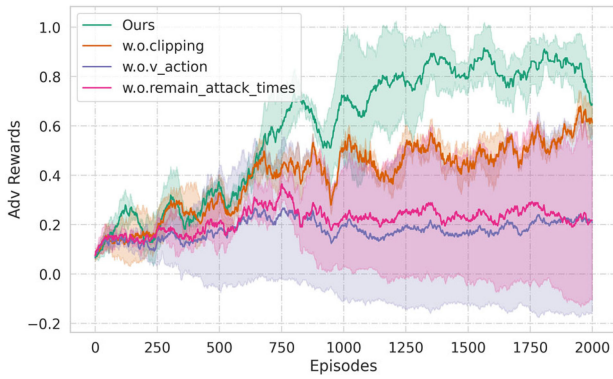


Fig. 7. Impact of each component on the performance of our attack method.

H. Discussion

This section presents a summary of how our method addresses the key questions RQ1–RQ5 and highlights its advantages.

- 1) *Response to RQ1:* Our method achieves a competitive CR with fewer attack steps in both unprotected left-turn and on-ramp merging scenarios, demonstrating superior AE.
- 2) *Response to RQ2:* Our attack method is effective against both standard RL and the latest state-of-the-art Safe RL and Robust RL approaches. In terms of attack performance, our algorithm achieves superior AE compared to UA and RA. Additionally, our method even outperforms the strong baseline AMA, as the stochastic nature of the perturbation generation process enhances exploration.
- 3) *Response to RQ3:* Our experiments show that once Γ increases beyond a certain threshold, it no longer affects ANA but instead accelerates the training process.
- 4) *Response to RQ4:* Our experiments demonstrate that the trained adversary remains effective even when the environment changes, such as variations in traffic density, showcasing its robustness.
- 5) *Response to RQ5:* The experimental results show that additional attack-related information is crucial for the adversary in learning effective attack policies. Meanwhile, trajectory clipping significantly accelerates the training process.

VI. CONCLUSION AND FUTURE WORK

This article proposes a novel adversarial attack method to conduct a stealthy and efficient attack against DRL-based autonomous driving agents. Our attack method is based on DRL, allowing it to learn the optimal attack policy in a domain-independent and self-learning manner. We validate its effectiveness across various state-of-the-art DRL algorithms, including vanilla RL, safe RL, and robust RL. Moreover, we compare our method with existing attack methods. Experimental results show that our attack method achieves over 90% CRs in most scenarios within three attack steps on average, improving AE by at least 105%.

Building on our current findings, several promising directions emerge for further investigation.

- 1) Developing more efficient perturbation generation methods is expected to further enhance the effectiveness of our method.
- 2) Exploring the adaptability of our method to different reward functions will provide valuable insights into its generalization.
- 3) Extensive experiments across diverse autonomous driving scenarios and against a broader range of potential defense strategies would more rigorously demonstrate the robustness of our method.
- 4) Investigating effective defense approaches against our attack method will contribute to improving the robustness of autonomous driving systems. Exploring equilibrium policies between attack and defense within a game-theoretic framework offers a promising research direction.
- 5) While our method achieves greater stealth by reducing attack frequency, quantitative evaluation using formal metrics (e.g., relative detection rates compared to high-frequency attacks) would provide more rigorous validation.

REFERENCES

- [1] “What might be the economic implications of autonomous vehicles?” Economics Observatory. Accessed: Dec. 3, 2024. [Online]. Available: <https://www.economicsobservatory.com/what-might-be-the-economic-implications-of-autonomous-vehicles>
- [2] X. Yan, Z. Zou, S. Feng, H. Zhu, H. Sun, and H. X. Liu, “Learning naturalistic driving environment with statistical realism,” *Nat. Commun.*, vol. 14, no. 1, p. 2037, Apr. 2023, doi: [10.1038/s41467-023-37677-5](https://doi.org/10.1038/s41467-023-37677-5).
- [3] M. Al-Sharman, R. Dempster, M. A. Daoud, M. Nasr, D. Rayside, and W. Melek, “Self-learned autonomous driving at unsignalized intersections: A hierarchical reinforced learning approach for feasible decision-making,” *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 11, pp. 12345–12356, Nov. 2023, doi: [10.1109/TITS.2023.3285440](https://doi.org/10.1109/TITS.2023.3285440).
- [4] J. Hu et al., “A survey of decision-making and planning methods for self-driving vehicles,” *Front. Neurobot.*, vol. 19, Feb. 2025, Art. no. 1451923, doi: [10.3389/fnbot.2025.1451923](https://doi.org/10.3389/fnbot.2025.1451923).
- [5] Z. Cao, S. Xu, X. Jiao, H. Peng, and D. Yang, “Trustworthy safety improvement for autonomous driving using reinforcement learning,” *Transp. Res. C, Emerg. Technol.*, vol. 138, May 2022, Art. no. 103656, doi: [10.1016/j.trc.2022.103656](https://doi.org/10.1016/j.trc.2022.103656).
- [6] K. Yang, X. Tang, S. Qiu, S. Jin, Z. Wei, and H. Wang, “Towards robust decision-making for autonomous driving on highway,” *IEEE Trans. Veh. Technol.*, vol. 72, no. 9, pp. 11251–11263, Sep. 2023, doi: [10.1109/TVT.2023.3268500](https://doi.org/10.1109/TVT.2023.3268500).
- [7] H. Deng, Y. Zhao, Q. Wang, and A.-T. Nguyen, “Deep reinforcement learning based decision-making strategy of autonomous vehicle in highway uncertain driving environments,” *Autom. Innov.*, vol. 6, no. 3, pp. 438–452, Aug. 2023, doi: [10.1007/s42154-023-00231-6](https://doi.org/10.1007/s42154-023-00231-6).
- [8] D. Chen et al., “Deep multi-agent reinforcement learning for highway on-ramp merging in mixed traffic,” *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 11, pp. 11623–11638, Nov. 2023, doi: [10.1109/TITS.2023.3285442](https://doi.org/10.1109/TITS.2023.3285442).
- [9] L. Liu, X. Li, Y. Li, J. Li, and Z. Liu, “Reinforcement learning-based multi-lane cooperative control for on-ramp merging in mixed-autonomy traffic,” *IEEE Internet Things J.*, vol. 11, no. 24, pp. 39809–39819, Dec. 2024, doi: [10.1109/IIOT.2024.3447039](https://doi.org/10.1109/IIOT.2024.3447039).
- [10] G. Li, W. Zhou, S. Lin, S. Li, and X. Qu, “On-ramp merging for highway autonomous driving: An application of a new safety indicator in deep reinforcement learning,” *Autom. Innov.*, vol. 6, no. 3, pp. 453–465, Aug. 2023, doi: [10.1007/s42154-023-00235-2](https://doi.org/10.1007/s42154-023-00235-2).
- [11] C.-J. Hoel, K. Wolff, and L. Laine, “Ensemble quantile networks: Uncertainty-aware reinforcement learning with applications in autonomous driving,” *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 6, pp. 6030–6041, Jun. 2023, doi: [10.1109/TITS.2023.3251376](https://doi.org/10.1109/TITS.2023.3251376).
- [12] E. Zhang, R. Zhang, and N. Masoud, “Predictive trajectory planning for autonomous vehicles at intersections using reinforcement learning,” *Transp. Res. C, Emerg. Technol.*, vol. 149, Apr. 2023, Art. no. 104063, doi: [10.1016/j.trc.2023.104063](https://doi.org/10.1016/j.trc.2023.104063).

- [13] W. Ding, L. Shi, Y. Chi, and D. Zhao, "Seeing is not believing: Robust reinforcement learning against spurious correlation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, 2024, pp. 1–36.
- [14] P. Buddareddygar, T. Zhang, Y. Yang, and Y. Ren, "Targeted attack on deep RL-based autonomous driving with learned visual patterns," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2022, pp. 10571–10577, doi: [10.1109/ICRA46639.2022.9811574](https://doi.org/10.1109/ICRA46639.2022.9811574).
- [15] A. Sharif and D. Marijan, "Adversarial deep reinforcement learning for improving the robustness of multi-agent autonomous driving policies," in *Proc. 29th Asia-Pacific Softw. Eng. Conf. (APSEC)*, Dec. 2022, pp. 61–70, doi: [10.1109/APSEC57359.2022.00018](https://doi.org/10.1109/APSEC57359.2022.00018).
- [16] A. D. M. Ibrahim, M. Hussain, and J.-E. Hong, "Deep learning adversarial attacks and defenses in autonomous vehicles: A systematic literature review from a safety perspective," *Artif. Intell. Rev.*, vol. 58, no. 1, p. 28, Nov. 2024, doi: [10.1007/s10462-024-1014-8](https://doi.org/10.1007/s10462-024-1014-8).
- [17] X. He, W. Huang, and C. Lv, "Trustworthy autonomous driving via defense-aware robust reinforcement learning against worst-case observational perturbations," *Transp. Res. C, Emerg. Technol.*, vol. 163, Jun. 2024, Art. no. 104632, doi: [10.1016/j.trc.2024.104632](https://doi.org/10.1016/j.trc.2024.104632).
- [18] Y. Deng, X. Zheng, T. Zhang, C. Chen, G. Lou, and M. Kim, "An analysis of adversarial attacks and defenses on autonomous driving models," in *Proc. IEEE Int. Conf. Pervasive Comput. Commun. (PerCom)*, Mar. 2020, pp. 1–10, doi: [10.1109/PerCom45495.2020.9127389](https://doi.org/10.1109/PerCom45495.2020.9127389).
- [19] X. He, B. Lou, H. Yang, and C. Lv, "Robust decision making for autonomous vehicles at highway on-ramps: A constrained adversarial reinforcement learning approach," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 4, pp. 4103–4113, Apr. 2023, doi: [10.1109/TITS.2022.3229518](https://doi.org/10.1109/TITS.2022.3229518).
- [20] C. Wang and N. Aouf, "Explainable deep adversarial reinforcement learning approach for robust autonomous driving," *IEEE Trans. Intell. Veh.*, early access, Mar. 19, 2024, doi: [10.1109/TIV.2024.3379367](https://doi.org/10.1109/TIV.2024.3379367).
- [21] X. He, H. Yang, Z. Hu, and C. Lv, "Robust lane change decision making for autonomous vehicles: An observation adversarial reinforcement learning approach," *IEEE Trans. Intell. Veh.*, vol. 8, no. 1, pp. 184–193, Jan. 2023, doi: [10.1109/TIV.2022.3165178](https://doi.org/10.1109/TIV.2022.3165178).
- [22] X. Cai, X. Bai, Z. Cui, P. Hang, H. Yu, and Y. Ren, "Adversarial stress test for autonomous vehicle via series reinforcement learning tasks with reward shaping," *IEEE Trans. Intell. Veh.*, early access, Jun. 25, 2024, doi: [10.1109/TIV.2024.3418887](https://doi.org/10.1109/TIV.2024.3418887).
- [23] A. Kulkarni, S. Zhang, and M. Behl, "CRASH: Challenging reinforcement-learning based adversarial scenarios for safety hardening," Nov. 2024, *arXiv:2411.16996*.
- [24] X. Ma, K. Driggs-Campbell, and M. J. Kochenderfer, "Improved robustness and safety for autonomous vehicle control with adversarial reinforcement learning," in *Proc. IEEE Intell. Veh. Symp.*, Jun. 2018, pp. 1665–1671, doi: [10.1109/IVS.2018.8500450](https://doi.org/10.1109/IVS.2018.8500450).
- [25] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," Aug. 2017, *arXiv:1707.06347*.
- [26] P. A. Lopez et al., "Microscopic Traffic Simulation using SUMO," in *Proc. 21st Int. Conf. Intell. Transp. Syst. (ITSC)*, Nov. 2018, pp. 2575–2582, doi: [10.1109/ITSC.2018.8569938](https://doi.org/10.1109/ITSC.2018.8569938).
- [27] J. Wu, C. Huang, H. Huang, C. Lv, Y. Wang, and F.-Y. Wang, "Recent advances in reinforcement learning-based autonomous driving behavior planning: A survey," *Transp. Res. C, Emerg. Technol.*, vol. 164, Jul. 2024, Art. no. 104654, doi: [10.1016/j.trc.2024.104654](https://doi.org/10.1016/j.trc.2024.104654).
- [28] Z. Huang, J. Wu, and C. Lv, "Efficient deep reinforcement learning with imitative expert priors for autonomous driving," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 10, pp. 7391–7403, Oct. 2023, doi: [10.1109/TNNLS.2022.3142822](https://doi.org/10.1109/TNNLS.2022.3142822).
- [29] Z. Li, F. Nie, Q. Sun, F. Da, and H. Zhao, "Boosting offline reinforcement learning for autonomous driving with hierarchical latent skills," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2024, pp. 18362–18369, doi: [10.1109/ICRA57147.2024.10611197](https://doi.org/10.1109/ICRA57147.2024.10611197).
- [30] F. Dang, D. Chen, J. Chen, and Z. Li, "Event-triggered model predictive control with deep reinforcement learning for autonomous driving," *IEEE Trans. Intell. Veh.*, vol. 9, no. 1, pp. 459–468, Jan. 2024, doi: [10.1109/TIV.2023.3329785](https://doi.org/10.1109/TIV.2023.3329785).
- [31] L. Chen, P. Wu, K. Chitta, B. Jaeger, A. Geiger, and H. Li, "End-to-end autonomous driving: Challenges and frontiers," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 10164–10183, Dec. 2024, doi: [10.1109/TPAMI.2024.3435937](https://doi.org/10.1109/TPAMI.2024.3435937).
- [32] Y.-C. Lin, Z.-W. Hong, Y.-H. Liao, M.-L. Shih, M.-Y. Liu, and M. Sun, "Tactics of adversarial attack on deep reinforcement learning agents," in *Proc. 26th Int. Joint Conf. Artif. Intell.*, 2017, pp. 3756–3762.
- [33] X. Li, Y. Li, Z. Feng, Z. Wang, and Q. Pan, "ATS-O2A: A state-based adversarial attack strategy on deep reinforcement learning," *Comput. Secur.*, vol. 129, Jun. 2023, Art. no. 103259, doi: [10.1016/j.cose.2023.103259](https://doi.org/10.1016/j.cose.2023.103259).
- [34] J. Sun et al., "Stealthy and efficient adversarial attacks against deep reinforcement learning," in *Proc. AAAI Conf. Artif. Intell.*, vol. 34, Apr. 2020, pp. 1–10, doi: [10.1609/aaai.v34i04.6047](https://doi.org/10.1609/aaai.v34i04.6047).
- [35] K. Mo, W. Tang, J. Li, and X. Yuan, "Attacking deep reinforcement learning with decoupled adversarial policy," *IEEE Trans. Dependable Secure Comput.*, vol. 20, no. 1, pp. 758–768, Jan./Feb. 2023, doi: [10.1109/TDSC.2022.3143566](https://doi.org/10.1109/TDSC.2022.3143566).
- [36] Keen Security Lab, "Experimental security research of Tesla autopilot," Tencent, Shenzhen, China, Whitepaper, 2019. [Online]. Available: https://keenlab.tencent.com/en/whitepapers/Experimental_Security_Research_of_Tesla_Autopilot.pdf
- [37] "Berlin researchers hacked tesla autopilot to unlock 'Elon mode'." 2024. [Online]. Available: <https://cybernews.com/tech/berlin-researchers-hacked-tesla-autopilot/>
- [38] Y. Cao et al., "Adversarial sensor attack on LiDAR-based perception in autonomous driving," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, Nov. 2019, pp. 2267–2281, doi: [10.1145/3319535.3339815](https://doi.org/10.1145/3319535.3339815).
- [39] C. Gongye, Y. Luo, X. Xu, and Y. Fei, "Side-Channel-Assisted Reverse-Engineering of Encrypted DNN hardware accelerator IP and attack surface exploration," in *Proc. IEEE Symp. Secur. Privacy*, May 2024, pp. 4678–4695, doi: [10.1109/SP54263.2024.00001](https://doi.org/10.1109/SP54263.2024.00001).
- [40] Y. Ren, H. Zhang, L. Du, Z. Zhang, J. Zhang, and H. Li, "Stealthy black-box attack with dynamic threshold against MARL-based traffic signal control system," *IEEE Trans. Ind. Inf.*, vol. 20, no. 10, pp. 12021–12031, Oct. 2024, doi: [10.1109/TII.2024.3413356](https://doi.org/10.1109/TII.2024.3413356).
- [41] A. Berdich and B. Groza, "Cyberattacks on adaptive cruise controls and emergency braking systems: adversary models, impact assessment, and countermeasures," *IEEE Trans. Reliab.*, vol. 73, no. 2, pp. 1216–1230, Jun. 2024, doi: [10.1109/TR.2024.3373810](https://doi.org/10.1109/TR.2024.3373810).
- [42] H. Sun et al., "CCID-CAN: Cross-chain intrusion detection on CAN bus for autonomous vehicles," *IEEE Internet Things J.*, vol. 11, no. 15, pp. 26146–26159, Aug. 2024, doi: [10.1109/JIOT.2024.3393122](https://doi.org/10.1109/JIOT.2024.3393122).
- [43] J. Fan, X. Chang, J. Mišić, V. B. Mišić, T. Yang, and Y. Gong, "Energy-constrained safe path planning for UAV-assisted data collection of mobile IoT devices," *IEEE Internet Things J.*, vol. 11, no. 24, pp. 39971–39983, Dec. 2024, doi: [10.1109/JIOT.2024.3448537](https://doi.org/10.1109/JIOT.2024.3448537).
- [44] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," in *Proc. Int. Conf. Mach. Learn.*, Jul. 2018, pp. 1861–1870.
- [45] S. Fujimoto, H. Hoof, and D. Meger, "Addressing function approximation error in actor-critic methods," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, Jul. 2018, pp. 1587–1596.
- [46] X. He et al., "Fear-neuro-inspired reinforcement learning for safe autonomous driving," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 1, pp. 267–279, Jan. 2024, doi: [10.1109/TPAMI.2023.3322426](https://doi.org/10.1109/TPAMI.2023.3322426).
- [47] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-baselines3: Reliable reinforcement learning implementations," *J. Mach. Learn. Res.*, vol. 22, no. 268, pp. 1–8, 2021.
- [48] A. Kurakin, I. Goodfellow, and S. Bengio, "Adversarial examples in the physical world," Feb. 2017, *arXiv:1607.02533*.
- [49] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *Proc. Int. Conf. Learn. Represent.*, 2018, pp. 1–28.



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